`Recurrence` is defined in Unix cron syntax format as shown here:

```
* * * * *
| | | | |
| | | | +- day of week (0 - 6) (0 Sunday)
| | | +--- month (1 - 12)
| | +----- day of month (1 - 31)
| +------- hour (0 - 23)
+--------- min (0 - 59)
```

We recommend using scheduled scaling actions whenever your infrastructure's capacity requirements are predictable—for example, an internal system used during work hours only, or a marketing action planned for a certain time.

17.2.2 Scaling based on CloudWatch metrics

Predicting the future is hard. Traffic will increase or decrease beyond known patterns from time to time. For example, if an article published on your blogging platform is heavily shared through social media, you need to be able to react to unplanned load changes and scale the number of EC2 instances.

You can adapt the number of EC2 instances to handle the current workload using CloudWatch alarms and scaling policies. CloudWatch helps monitor virtual machines and other services on AWS. Typically, services publish usage metrics to CloudWatch, helping you to evaluate the available capacity. The types of scaling policies follow:

1. *Step scaling*—Allows more advanced scaling because multiple scaling adjustments are supported, depending on how much the threshold you set has been exceeded.
2. *Target tracking*—Frees you from defining scaling steps and thresholds. You need only to define a target (such as CPU utilization of 70%), and the number of EC2 instances is adjusted accordingly.
3. *Predictive scaling*—Uses machine learning to predict load. It works best for cyclical traffic and recurring workload patterns (see "Predictive Scaling for Amazon EC2 Auto Scaling" at http://mng.bz/RvYO to learn more).
4. *Simple scaling*—A legacy option that was replaced with step scaling.

All of the scaling policies use metrics and alarms to scale the number of EC2 instances based on the current workload. As shown in figure 17.4, the virtual machines publish metrics to CloudWatch constantly. A CloudWatch alarm monitors one of these metrics and triggers a scaling action if the defined threshold is reached.